A Vanilla zkVM Architecture

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This document is a reference example illustrating the expected scope and level of rigor. It is not meant to describe a zkVM that is implementable and efficient, and the described zkVM may not even be sound. Submitted documents should provide additional detail where applicable.

Overview

This document describes a sample *vanilla zkVM* architecture designed according to the W1 deliverable of the EF zkVM whitepaper guidelines[1]. The architecture proves execution of a fixed RV+ (extended RISC-V) guest program using:

- Segmented execution traces
- Specialized constraint systems ("chips")
- A system of interaction buses
- LogUp arguments for buses consistency
- WHIR-based STARK proofs
- Recursive aggregation of segment proofs

After execution, the trace is divided into segments. Each segment is proven by four inner circuits tied together by a shared bus, whose proofs are verified by a single segment circuit. Segment proofs are then recursively aggregated in a binary tree until a single proof attests to correctness of the entire program execution.

1 Execution Model

We define how a program is executed.

1.1 Program, Execution, and VM State

A **program** $\mathcal{P}$ is a fixed RV+ binary, compiled from an Ethereum State Transition function. Its inputs $\mathcal{I}$ are derived from the Ethereum state (*initial state*), and the execution witness $\mathcal{W}$, and its outputs $\mathcal{O}$ are the final Ethereum state (*final state*).

The program $\mathcal{P}$ operates on the **VM state** S, defined as concatenation of the following components:

[1] `https://zkevm.ethereum.foundation/blog/cryptography-research-update`

- $\texttt{pc}$: 32-bit program counter
- $\texttt{regs}[0..k-1]$: list of k 32-bit register values
- $\texttt{mem}[]$: byte-addressed writable memory

We separately denote the code of the program by $\texttt{code}$. It is immutable and read-only, and is not part of the VM state.

A single instruction step of the program execution is defined as a transition from one VM state to another according to the semantics of the instruction op retrieved from $\texttt{code}$ at the program counter $\texttt{pc}_1$. This is denoted as follows:

$$S := (\texttt{pc}_1, \texttt{regs}_1, \texttt{mem}_1) \xrightarrow{\text{op}:=\texttt{code}[\texttt{pc}_1]} S' := (\texttt{pc}_2, \texttt{regs}_2, \texttt{mem}_2) \tag{1}$$

For example, if op is an addition instruction, which adds the first two registers and stores in the third register, then the semantics of the step is defined as

$$\texttt{pc}_2 = \texttt{pc}_1 + 1, \tag{2}$$
$$\texttt{regs}_2 = \texttt{regs}_1 \text{ except that } \texttt{regs}_2[2] := \texttt{regs}_1[0] + \texttt{regs}_1[1], \tag{3}$$
$$\texttt{mem}_2 = \texttt{mem}_1. \tag{4}$$

We will use the notation $(\texttt{pc}_1, \texttt{regs}_1, \texttt{mem}_1) \to (\texttt{pc}_2, \texttt{regs}_2, \texttt{mem}_2)$ to denote that S transitions to S' by executing instruction $\text{op} := \texttt{code}[\texttt{pc}_1]$.

An **execution trace** for program $\texttt{code}$ applied to inputs $\mathcal{I}$ is a sequence of VM states

$$\mathcal{T}: \quad S_0 \to S_1 \to \cdots \to S_T$$

where each step corresponds to execution of one instruction.

A **zkVM proof** for $(\mathcal{P}, \mathbb{E}, \mathbb{E}')$, where $\mathbb{E}$ and $\mathbb{E}'$ are the initial and final Ethereum states, is a SNARK that proves the following:

$$\begin{cases} S_0 \to \cdots \to S_T; \\ S_0 := (\texttt{pc}_0 := 0, \texttt{regs}, \texttt{mem}) \text{ where } \texttt{regs}, \texttt{mem} \text{ are derived from } \mathbb{E}; \\ S_T := (\texttt{pc}_T, \texttt{regs}', \texttt{mem}') \text{ where } \texttt{regs}', \texttt{mem}' \text{ are derived from } \mathbb{E}'; \end{cases}$$

1.2 Supported Instructions

The zkVM supports the RV+ instruction set, which is an extension of the RV32IM instruction set. Concretely, all operations:

- conform to the state transition (1), i.e. transform $(\texttt{pc}_1, \texttt{regs}_1, \texttt{mem}_1)$ to $(\texttt{pc}_2, \texttt{regs}_2, \texttt{mem}_2)$ according to the semantics of the instruction op retrieved from $\texttt{code}$ at the program counter $\texttt{pc}_1$.
- are all standard RISC-V instructions[2] (executed according to the standard RISC-V semantics) or additional instructions (precompiles) for efficient execution of certain operations — concretely, Keccak and Poseidon.

[2] https://riscv.org/technical/specifications/

For our purpose the exact semantics of the instructions is not important, and we treat them as black boxes with their own predicates modeling the correctness of the execution.

Concretely, for given instruction *op*, the predicate φ_{op} has the semantics

$$\varphi_{op}(\mathtt{pc}_1, \mathtt{regs}_1, \mathtt{mem}_1, \mathtt{pc}_2, \mathtt{regs}_2, \mathtt{mem}_2) = \text{TRUE} \iff (\mathtt{pc}_1, \mathtt{regs}_1, \mathtt{mem}_1) \xrightarrow{\text{op}} (\mathtt{pc}_2, \mathtt{regs}_2, \mathtt{mem}_2) \wedge \mathtt{code}[\mathtt{pc}_1] = op \quad (5)$$

For instance, if *op* is an ADD instruction, then $\mathtt{mem}_1 = \mathtt{mem}_2$, $\mathtt{pc}_1 + 1 = \mathtt{pc}_2$ and the registers change according to the ADD instruction.

Exceptions are memory operations, which we assume have the following syntax and semantics:

$$\text{read}: \ (\mathtt{regs}_1 := (addr, *, \ldots), \mathtt{mem}) \to (\mathtt{regs}_2 := (addr, x, \ldots), \mathtt{mem}): \qquad \mathtt{mem}[addr] = x, \quad (6)$$

$$\text{write}: \ (\mathtt{regs}_1 := (addr, x, \ldots), \mathtt{mem}) \to (\mathtt{regs}_2 := (addr, x, \ldots), \mathtt{mem}'): \qquad \mathtt{mem}'[i] = \begin{cases} x & \text{if } i = addr, \\ \mathtt{mem}[i] & \text{otherwise.} \end{cases} \quad (7)$$

2 Segmentation

We now explain how execution is split into segments.

2.1 Execution Segments

After execution, the execution trace $\mathcal{T}$ is divided into segments $\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_m$ of exactly N_{seg} instructions[3] each (cf. Fig. 1). Each segment $\mathcal{G}_i$ corresponds to a contiguous subsequence of the execution trace, from S_{start_i} to S_{end_i}, and is proven by a set of separate proofs. The segmentation of the execution trace into consecutive segments allows reducing the proof of the entire execution to proving the validity of each segment and consistency between them. The global consistency argument asserts that the segments are consistent with each other and with the public inputs/outputs:

$$S_{end_i} = S_{start_{i+1}} \text{ for all } i \in [m-1], \quad S_{start_1} = S_0, \quad S_{end_m} = S_T.$$

2.2 Segment layout

Each segment is represented in an equivalent form as a collection of one trace and several *chips*, listed below (cf. Fig. 2).

Segment trace It consists of all operations of the execution trace within the segment and state variables, including the program counter, register values, memory state, etc. It is a collection of N_{seg} rows, each corresponding to one instruction step from the segment. The segment trace does not enforce the semantics of all instructions, but only a subset of them, and the rest of the instructions are proven in separate chips and their consistency with the segment trace is enforced by *consistency arguments*.

Keccak chip It consists of all Keccak precompile calls in the segment, and their inputs/outputs.

[3]The instruction set has a no-op instruction to pad the segment to the full length. The parameter N_{seg} is assumed to be hardcoded and fixed, and known to the verifier.

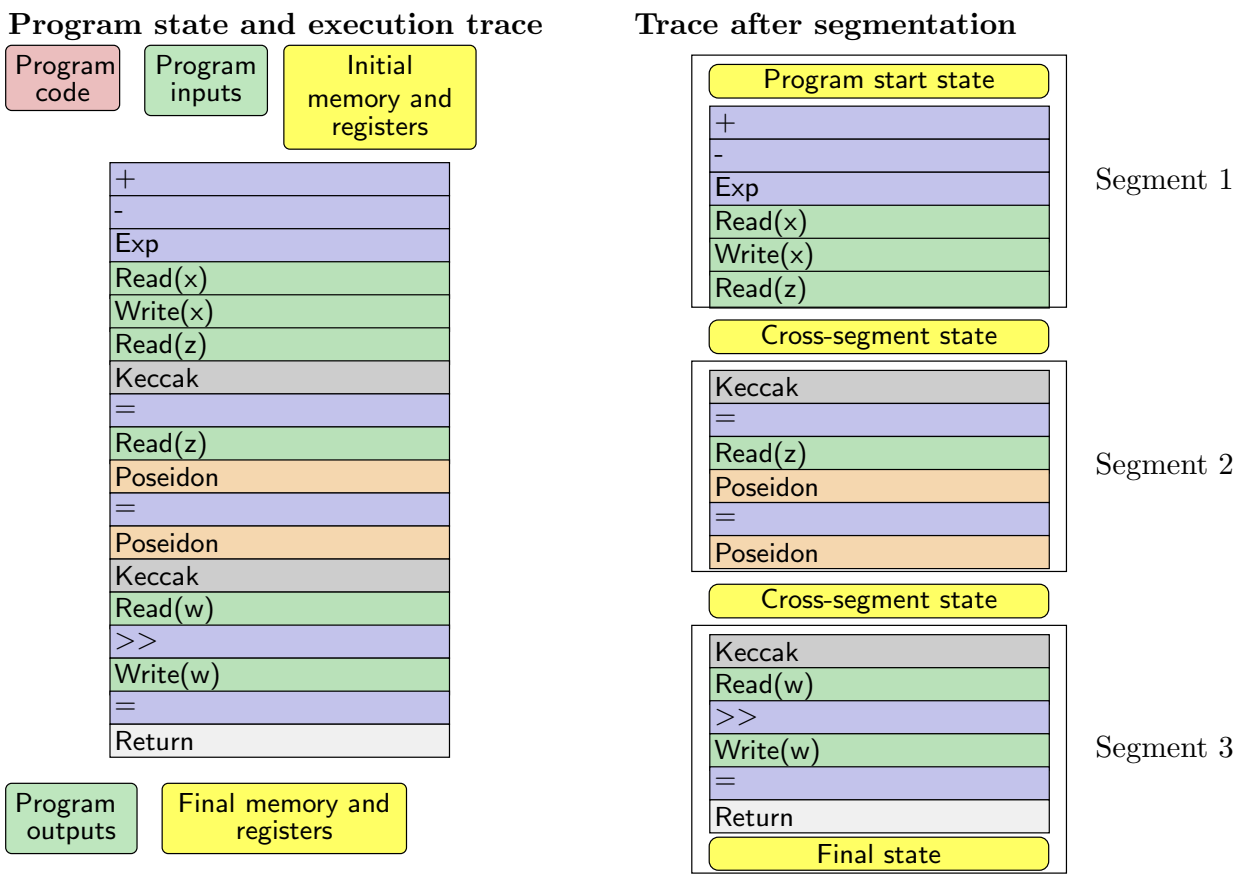

Figure 1: Segmentation of the execution trace into 3 segments

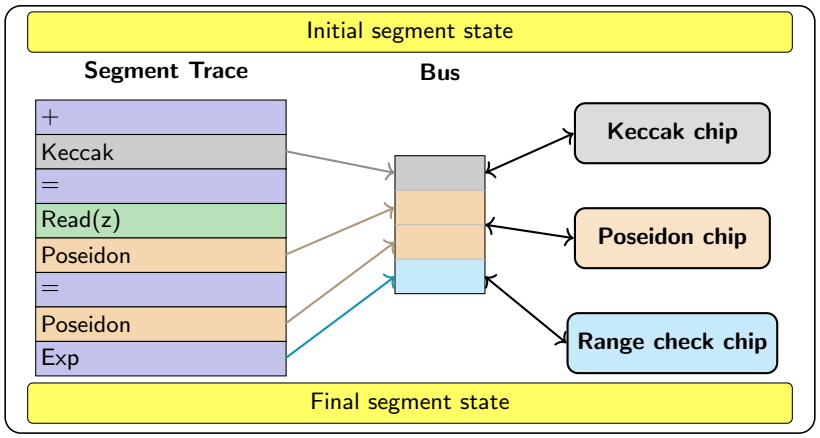

Figure 2: Segment structure: segment trace connects to the bus which then connects to chips.

Poseidon chip It consists of all Poseidon precompile calls in the segment, and their inputs/outputs.

Range check chip It consists of all range check statements occurring in the segment.

Local consistency argument

Teams should describe how their bus consistency argument works in terms of lookups used and how they integrate into the overall proof system.

The consistency arguments enforce that the operations proven in the chips are consistent with the segment trace. This is done using the concept of a *bus* connecting the segment trace and the chips, and lookup arguments to enforce consistency of the bus values with the segment trace and the chip traces.

3 Correct execution of a program

In this section, we define via predicates what it means that a program is correctly executed. We will use these predicates in Section 4 to define the relations that the zkVM proves.

3.1 Predicates for individual operations

Teams should list all operations supported by their zkVM and define the corresponding predicates based on their architecture.

Each operation is treated as a black box with its own predicate φ_{op} (cf. (5)). Throughout this section, subscripts 1 and 2 denote states before and after an operation. We also denote $S_1 := (\mathtt{pc}_1, \mathtt{regs}_1, \mathtt{mem}_1)$ and $S_2 := (\mathtt{pc}_2, \mathtt{regs}_2, \mathtt{mem}_2)$ as the states before and after the operation.

We group operations as follows:

- **Memory operations.** Read: $\varphi_{\text{read}}(S_1, S_2)$ — does not modify memory. Write: $\varphi_{\text{write}}(S_1, S_2)$ — may modify memory.
- **Precompiles.** Keccak: $\varphi_{\text{keccak}}(S_1, S_2)$. Poseidon: $\varphi_{\text{poseidon}}(S_1, S_2)$.

- **Arithmetic without range checks.** $\mathrm{op}_1, \ldots, \mathrm{op}_{10}$ with predicates $\varphi_{\mathrm{op_i}}(S_1, S_2)$.
- **Arithmetic with range checks.** $\mathrm{op}_{11}, \ldots, \mathrm{op}_{20}$ with predicates $\varphi_{\mathrm{op_i}}(S_1, S_2)$, each decomposable as $\varphi'_{\mathrm{op_i}}(S_1, S_2) \wedge \varphi_{\mathrm{range}}(S_1)$ where the constants defining the range check are always in the last two registers.

All operations except read and write do not read or write memory, so we may omit these arguments from their predicates in the future text for brevity.

3.2 Correct execution of a single step

Teams should define the step predicate for their architecture, specifying how the correct operation is selected and verified at each step.

A single step executes the instruction at the current program counter. The step predicate selects the matching operation via a disjunction over all opcodes:

$$\varphi_{step}(S_1, S_2) := \bigvee_{op \in \mathrm{operations}} \varphi_{op}(S_1, S_2)$$

where $S_1 := (\mathtt{pc}_1, \mathtt{regs}_1, \mathtt{mem}_1)$ and $S_2 := (\mathtt{pc}_2, \mathtt{regs}_2, \mathtt{mem}_2)$ are the states before and after the step.

3.3 Delegating expensive operations using a Bus

Teams should describe how expensive operations (e.g., precompiles, range checks) are delegated to a bus or similar mechanism in their architecture.

Range checks and precompile calls are expensive to prove inline. Instead, we collect them in a shared *bus* $\mathcal{B}$ and verify separately, proving the consistency along the way. The *bus* $\mathcal{B}$ is a collection of all precompile calls and range checks within a segment. That is, it has the following structure:

$$\mathcal{B} = \{(\underbrace{\mathrm{op}^{(i)}}_{\in\{\mathrm{keccak,poseidon}\}}, S_1^{(i)}, S_2^{(i)})\}_{i=1}^{b} \cup \{(\mathrm{range}, S^{(j)})\}_{j=1}^{r}$$

where b is the number of precompile calls and r is the number of range checks.

Now we define auxiliary predicates that filter the bus by operation type and assert correctness of each entry: $\varphi_{\mathrm{keccak}}(\mathcal{B})$ checks all Keccak entries, $\varphi_{\mathrm{poseidon}}(\mathcal{B})$ checks all Poseidon entries, and $\varphi_{\mathrm{range}}(\mathcal{B})$ checks all range-check entries. Concretely:

$$\varphi_{\mathrm{keccak}}(\mathcal{B}) := \bigwedge_{(\mathrm{keccak}, S_1, S_2) \in \mathcal{B}} \varphi_{\mathrm{keccak}}(S_1, S_2)$$

$$\varphi_{\mathrm{poseidon}}(\mathcal{B}) := \bigwedge_{(\mathrm{poseidon}, S_1, S_2) \in \mathcal{B}} \varphi_{\mathrm{poseidon}}(S_1, S_2)$$

$$\varphi_{\mathrm{range}}(\mathcal{B}) := \bigwedge_{(\mathrm{range}, S) \in \mathcal{B}} \varphi_{\mathrm{range}}(S)$$

We now rewrite the step predicate (3.2) to separate inline verification from bus-deferred verification. The new form is equivalent[4]:

$$\varphi_{\mathrm{step}}(S_1, S_2, \mathcal{B}) := \varphi_{\mathrm{step,bus}}(S_1, S_2, \mathcal{B}) \wedge \varphi_{\mathrm{keccak}}(\mathcal{B}) \wedge \varphi_{\mathrm{poseidon}}(\mathcal{B}) \wedge \varphi_{\mathrm{range}}(\mathcal{B}) \tag{8}$$

[4]Formally the new predicate is even stronger as it also asserts the correctness of the operations in $\mathcal{B}$ not listed in the segment. We omit checking for such operations for efficiency.

where

$$\varphi_{\text{step,bus}}(S_1, S_2, \mathcal{B}) := \bigvee_{op \in \{\text{read,write,op}_1,\ldots,\text{op}_{10}\}} (\varphi_{\text{op}}(S_1, S_2)) \bigvee_{op \in \{\text{op}_{11},\ldots,\text{op}_{20}\}} \left(\varphi'_{\text{op}}(S_1, S_2) \wedge (\text{range}, S_1) \in \mathcal{B}\right) \bigvee_{op \in \{\text{Keccak,Poseidon}\}} ((op, S_1, S_2) \in \mathcal{B}) \quad (9)$$

Recall that $S_1 := (\texttt{pc}_1, \texttt{regs}_1, \texttt{mem}_1)$ and $S_2 := (\texttt{pc}_2, \texttt{regs}_2, \texttt{mem}_2)$. The predicate $\varphi_{\text{step,bus}}$ separates operations into three categories:

- **Memory ops and arithmetic without range checks** $(\text{read}, \text{write}, \text{op}_1, \ldots, \text{op}_{10})$ are verified inline via their predicates.
- **Arithmetic with range checks** $(\text{op}_{11}, \ldots, \text{op}_{20})$: the arithmetic logic φ_{op} is verified inline, but the range check is deferred by recording (range, S_1) in $\mathcal{B}$.
- **Precompiles** (Keccak, Poseidon) are fully deferred — the step only checks that the call appears in $\mathcal{B}$, while correctness will be verified by $\varphi_{\text{keccak}}(\mathcal{B})$, $\varphi_{\text{poseidon}}(\mathcal{B})$ in (8).

3.4 Committed memory

Teams should describe how memory is committed and how memory operations are verified in their architecture.

The full memory state is too large to pass around directly, so we replace it with the Merkle tree commitment Com. We define a *state with committed memory* $\hat{S} := (\texttt{pc}, \texttt{regs}, \text{Com})$, replacing the full memory in $S := (\texttt{pc}, \texttt{regs}, \texttt{mem})$ with its commitment.

The read operation predicate becomes

$$\widehat{\varphi}_{\text{read}}(\texttt{pc}_1, \texttt{regs}_1, \text{Com}_1, \texttt{pc}_2, \texttt{regs}_2, \text{Com}_2)$$

with $\text{Com}_1 = \text{Com}_2$, and the write operation predicate becomes

$$\widehat{\varphi}_{\text{write}}(\texttt{pc}_1, \texttt{regs}_1, \text{Com}_1, \texttt{pc}_2, \texttt{regs}_2, \text{Com}_2)$$

with additional semantics that the read and write operations are consistent with the commitments.

Replacing memory with commitments in (8) gives the step relation:

$$\widehat{\varphi}_{step}(\hat{S}_1, \hat{S}_2, \widehat{\mathcal{B}}) := \widehat{\varphi}_{step,bus}(\hat{S}_1, \hat{S}_2, \widehat{\mathcal{B}}) \wedge \widehat{\varphi}_{\text{keccak}}(\widehat{\mathcal{B}}) \wedge \widehat{\varphi}_{\text{poseidon}}(\widehat{\mathcal{B}}) \wedge \widehat{\varphi}_{\text{range}}(\widehat{\mathcal{B}}) \quad (10)$$

where $\widehat{\mathcal{B}}$ replaces the memory entries with committed memory (which is practically invisible as the bus operations do not use memory), $\widehat{\varphi}_{\text{keccak}}$, $\widehat{\varphi}_{\text{poseidon}}$, $\widehat{\varphi}_{\text{range}}$ are essentially unchanged as those relations do not use memory, and $\widehat{\varphi}_{step,bus}$ are defined analogously to $\varphi_{\text{step,bus}}$ but with all operation predicates replaced by their committed-memory versions:

$$\widehat{\varphi}_{step,bus}(\hat{S}_1, \hat{S}_2, \widehat{\mathcal{B}}) := \bigvee_{op \in \{\text{read,write,op}_1,\text{op}_2,\ldots,\text{op}_{10}\}} \left(\widehat{\varphi}_{op}(\hat{S}_1, \hat{S}_2)\right) \bigvee_{op \in \{\text{op}_{11},\text{op}_{12},\ldots,\text{op}_{20}\}} \left(\widehat{\varphi}'_{op}(\hat{S}_1, \hat{S}_2) \wedge (\text{range}, \hat{S}_1) \in \widehat{\mathcal{B}}\right) \bigvee_{op \in \{\text{Keccak,Poseidon}\}} \left((op, \hat{S}_1, \hat{S}_2) \in \widehat{\mathcal{B}}\right) \quad (11)$$

Note that (8) and (10) are no longer equivalent (in terms of soundness) as the second one uses committed memory instead of the actual memory.

4 Proof Architecture

We now define relations that are proven using individual SNARKs, and that recursively combine into the final proof.

4.1 Topology

The proof pipeline consists of multiple circuit types.

Teams should add footnotes to each circuit subsection below linking to the corresponding soundcalc entry for that circuit's parameters and security analysis.

Below we expose some aggregate information.

Circuit	**Purpose**	Relations
Inner segment circuits (leaf layer):		
`inner-step`	Segment trace	$\mathcal{R}_{0,\text{step}}$
`inner-keccak`	Keccak precompile	$\mathcal{R}_{0,\text{keccak}}$
`inner-poseidon`	Poseidon precompile	$\mathcal{R}_{0,\text{poseidon}}$
`inner-range`	Range checks	$\mathcal{R}_{0,\text{range}}$
`segment`	Segment proof	$\mathcal{R}_1$
`convert`	1-to-1 normalization	$\mathcal{R}_2$
`combine`	2-to-1 merge	$\mathcal{R}_3$
`embed`	final proof	$\mathcal{R}_4$

- *Inner segment circuits (leaf layer).* These four circuits jointly prove one segment; the `segment` circuit below verifies all four.
  - `inner-step`: Proves the segment trace execution (with precompiles and range checks excluded).
  - `inner-keccak`: Proves the Keccak precompile.
  - `inner-poseidon`: Proves the Poseidon precompile.
  - `inner-range`: Proves the range checks.
- `segment`: Verifies the four inner proofs, effectively proving one segment
- `convert`: A 1-to-1 recursion circuit that verifies a single `segment` proof.
- `combine`: A recursive 2-to-1 circuit that verifies two child proofs, each of which may be either a `convert` or a `combine` proof.
- `embed`: The circuit that produces the final proof.

We visualize the recursion topology in Fig. 3. Here, $\pi^{\mathbf{0}}$ denotes the four inner proofs (one each for $\mathcal{R}_{0,\text{step}}$, $\mathcal{R}_{0,\text{keccak}}$, $\mathcal{R}_{0,\text{poseidon}}$, $\mathcal{R}_{0,\text{range}}$), π^1 are proofs of $\mathcal{R}_1$ and thus the `segment` circuit. Similarly, π^2 proves the `convert` circuit, π^3 the `combine` circuit, and π^4 the `embed` circuit.

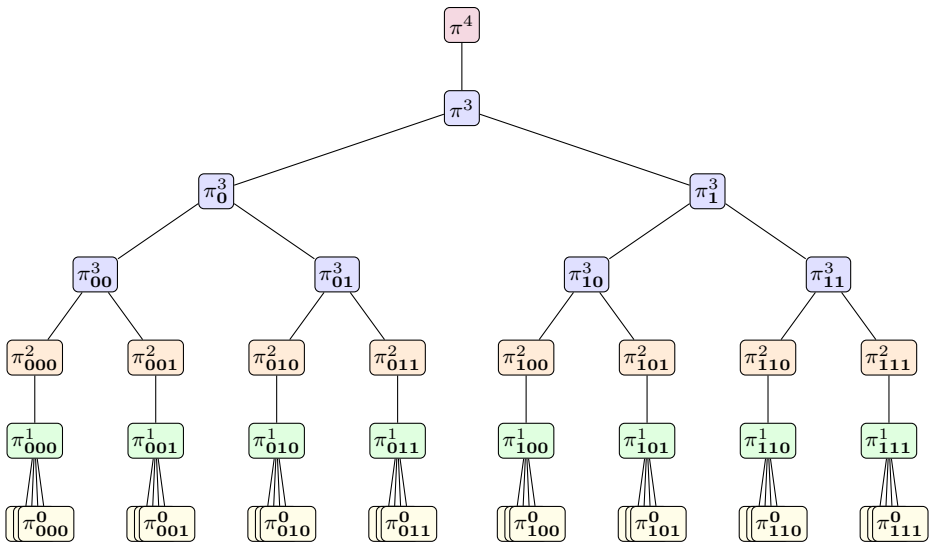

Figure 3: Recursion topology. Colors: `inner`, `segment`, `convert`, `combine`, `embed`.

4.2 `inner` circuits

To optimize the proof size and prover time, we introduce inner segment circuits that prove the correctness of the segment trace, of precompiles, and of range checks separately. As those circuits share the same bus, we commit to the latter to make the relations succinct. These circuits are then recursively combined to prove the correctness of the full segment.

The first relation $\mathcal{R}_{0,\text{step}}$ chains N_{seg} consecutive valid steps from $\hat{S}_{in}$ to $\hat{S}_{out}$, with the bus committed in the public input. The remaining three relations each assert correctness of all entries of a given type in the bus (Keccak, Poseidon, or range checks respectively), again with the bus committed[5] in the public input.

$$\mathcal{R}_{0,\text{step}} = \left(\left((\hat{S}_{in}, \hat{S}_{out}, \text{Com}_B);\ (\widehat{\mathcal{B}}, \{\hat{S}_i\}_{1 \le i \le N_{\text{seg}}+1}) \right) \;\middle|\; \begin{array}{c} \hat{S}_1 = \hat{S}_{in} \\ \hat{S}_{N_{\text{seg}}+1} = \hat{S}_{out} \\ \bigwedge_{i=1}^{N_{\text{seg}}} \widehat{\varphi}_{step,bus}(\hat{S}_i, \hat{S}_{i+1}, \widehat{\mathcal{B}}) = \text{TRUE} \\ \text{Com}_B = \text{Com}(\widehat{\mathcal{B}}) \end{array} \right) \tag{12}$$

$$\mathcal{R}_{0,\text{keccak}} = \left(\left((\text{Com}_B);\ (\widehat{\mathcal{B}}) \right) \;\middle|\; \begin{array}{c} \widehat{\varphi}_{\text{keccak}}(\widehat{\mathcal{B}}) = \text{TRUE} \\ \text{Com}_B = \text{Com}(\widehat{\mathcal{B}}) \end{array} \right) \tag{13}$$

$$\mathcal{R}_{0,\text{poseidon}} = \left(\left((\text{Com}_B);\ (\widehat{\mathcal{B}}) \right) \;\middle|\; \begin{array}{c} \widehat{\varphi}_{\text{poseidon}}(\widehat{\mathcal{B}}) = \text{TRUE} \\ \text{Com}_B = \text{Com}(\widehat{\mathcal{B}}) \end{array} \right) \tag{14}$$

$$\mathcal{R}_{0,\text{range}} = \left(\left((\text{Com}_B);\ (\widehat{\mathcal{B}}) \right) \;\middle|\; \begin{array}{c} \widehat{\varphi}_{\text{range}}(\widehat{\mathcal{B}}) = \text{TRUE} \\ \text{Com}_B = \text{Com}(\widehat{\mathcal{B}}) \end{array} \right) \tag{15}$$

The circuits are called, respectively, `inner-step`, `inner-keccak`, `inner-poseidon`, and `inner-range`.

Note that our writeup is a bit informal here, as we never formally defined Com. Teams should do that, as the security analysis later will require a specific binding notion of Com.

[5] Without loss of generality we assume that the same commitment scheme is used for the bus.

4.3 `segment` circuit

The `segment` circuit recursively verifies the four inner proofs, tying them together via a shared bus commitment Com_B. Let $\Pi_{0,j} = (\Pi_{0,j}.\mathsf{Prove}, \Pi_{0,j}.\mathsf{Verify})$ be the SNARK for $\mathcal{R}_{0,j}$, where $j \in \{\mathrm{step}, \mathrm{keccak}, \mathrm{poseidon}, \mathrm{range}\}$.

$$\mathcal{R}_1 = \left\{ (\hat{S}_{in}, \hat{S}_{out}); (\mathrm{Com}_B, \pi_{\mathrm{step}}, \pi_{\mathrm{keccak}}, \pi_{\mathrm{poseidon}}, \pi_{\mathrm{range}}) \;\middle|\; \begin{array}{c} \Pi_{0,\mathrm{step}}.\mathsf{Verify}((\hat{S}_{in},\ \hat{S}_{out},\ \mathrm{Com}_B), \pi_{\mathrm{step}}) = 1 \\ \Pi_{0,\mathrm{keccak}}.\mathsf{Verify}(\mathrm{Com}_B, \pi_{\mathrm{keccak}}) = 1 \\ \Pi_{0,\mathrm{poseidon}}.\mathsf{Verify}(\mathrm{Com}_B, \pi_{\mathrm{poseidon}}) = 1 \\ \Pi_{0,\mathrm{range}}.\mathsf{Verify}(\mathrm{Com}_B, \pi_{\mathrm{range}}) = 1 \end{array} \right\}$$

4.4 `convert` circuit

The `convert` circuit is a 1-to-1 recursion circuit that consumes a proof of $\mathcal{R}_1$ and verifies it. Its public inputs are the boundary states and the segment size; its witness is the proof π_1. To formally define the relation, let $\Pi_1 = (\Pi_1.\mathsf{Prove}, \Pi_1.\mathsf{Verify})$ be the STARK proving protocol for relation $\mathcal{R}_1$.

$$\mathcal{R}_2 = \left\{ \ ((\hat{S}_0,\ \hat{S}_N,\ N_{\mathrm{seg}});\ \pi_1) \;\middle|\; \Pi_1.\mathsf{Verify}((\hat{S}_0,\ \hat{S}_N), \pi_1) = 1 \ \right\}$$

Note that triples for which the last component is not N_{seg} are never in $\mathcal{R}_2$.

4.5 `combine` circuit

The `combine` circuit is a recursive 2-to-1 circuit that verifies two child proofs, each of which may be either a `convert` proof or another `combine` proof. It checks three things:

- Each child proof is valid under either $\Pi_2.\mathsf{Verify}$ or $\Pi_3.\mathsf{Verify}$, where $\Pi_i = (\Pi_i.\mathsf{Prove}, \Pi_i.\mathsf{Verify})$ is the STARK proving protocol for relation $\mathcal{R}_i$.
- The step counts satisfy $N_L + N_R = N$, with $N_L \geq 1$ and $N_R \geq 1$ to ensure well-founded recursion.
- State continuity: the witness $\hat{S}_N^L$ serves as both the final state of the left proof and the initial state of the right proof.

Since `combine` is self-recursive, repeated binary application naturally reduces any number of segment proofs into a single proof π_3 for the full execution.

For brevity, we write $\mathsf{V}_i(\mathsf{x}, \pi) := \Pi_i.\mathsf{Verify}(\mathsf{x}, \pi)$.

$$\mathcal{R}_3 = \left\{ ((\hat{S}_0^L,\ \hat{S}_N^R,\ N);\ (\pi^L,\ \pi^R,\ \hat{S}_N^L, N_L, N_R)) \;\middle|\; \begin{array}{c} \mathsf{V}_2((\hat{S}_0^L, \hat{S}_N^L, N_L),\ \pi^L) = 1 \ \vee\ \mathsf{V}_3((\hat{S}_0^L, \hat{S}_N^L, N_L),\ \pi^L) = 1 \\ \mathsf{V}_2((\hat{S}_N^L, \hat{S}_N^R, N_R),\ \pi^R) = 1 \ \vee\ \mathsf{V}_3((\hat{S}_N^L, \hat{S}_N^R, N_R),\ \pi^R) = 1 \\ N_L + N_R = N \\ N_L \geq 1,\ N_R \geq 1 \end{array} \right\}$$

where $\hat{S}_0^L$ and $\hat{S}_N^R$ are the public boundary states of the combined execution, π^L and π^R are the two child proofs (each either a `convert` or `combine` proof), and $\hat{S}_N^L$ is the intermediate boundary state provided as witness.

4.6 `embed` circuit

The `embed` circuit produces the final proof, asserting correctness of the entire program execution with public initial and final states. It assumes that the total step count T is known in advance to the verifier. Let $\Pi_3 = (\Pi_3.\mathsf{Prove}, \Pi_3.\mathsf{Verify})$ be the STARK proving protocol for relation $\mathcal{R}_3$.

$$\mathcal{R}_4 = \left\{ \ ((\hat{S}_0,\, \hat{S}_T);\ \pi_3) \ \middle|\ \Pi_3.\mathsf{Verify}((\hat{S}_0,\, \hat{S}_T,\, T),\, \pi_3) = 1 \ \right\}$$

where $\hat{S}_0$ and $\hat{S}_T$ are the initial and final states of the entire program execution.

5 Proof system instantiations

Teams should explain how the relations are proven using AIR / IOPs / poly-IOPs and the BCS transform. That is, teams should provide a direct reduction from proving a relation to a low-degree test with FRI or another protocol. The reductions should contain links to the circuits submitted to soundcalc. The purpose of the section is to make explicit how the relations from the previous section are proven.

In the previous sections, we have described how to arithmetize the correct RISC-V execution of a program by decomposing it into a collection of tables of field elements, each of which has algebraic constraints placed on its columns. Furthermore, we allow tables to "send" and "receive" messages with multiplicity through a collection of "interaction kinds". All tables are zero padded to have the same height, call it $\mathcal{N}_{row} = 2^{n_{row}}$. For the RISC-V circuits, we use $n_{row} = 22$. The reason for this padding is that it allows for the verifier to be described as a uniform arithmetic circuit, independent of the unpadded table heights. This naturally introduces some prover and verifier overhead; naively, this can be a very large overhead if one of the tables is wide and short, in which case the padding adds a substantial number of trace cells, but the Hypercube proof system has a number of mitigations to minimize this overhead. We will describe some of these mitigations below, especially because one of them is a cryptographic protocol invented for this particular zkVM context. First, though, let us describe protocol as if it were actually run on the padded tables:

1. Treat each column of each table as a multilinear polynomial in n_{row} variables. Use a batch multilinear polynomial commitment scheme to commit to these polynomials. Let W denote the total number of columns/multilinear polynomials.

2. Use GKR with n_{row} maybe ± 1 rounds to prove the LogUp argument equating the "send" and "receive" multisets across all tables. The output of GKR is a collection of W evaluation claims, one on each column, all at the same point η.

3. Use the point η as the randomness for the zerocheck sumcheck which proves constraint satisfaction. Batch the zerocheck sumcheck with the sumchecks which "prove" the evaluation claims from LogUp+GKR (more precisely, the sumchecks reduce evaluation claims at one point to evaluation claims at another). The output of this batched sumcheck is again a collection of W evaluation claims, not at a new point η'.

4. Use the batch multilinear polynomial commitment scheme to prove these evaluation claims.

If the prover actually materialized the full $W \times \mathcal{N}_{row}$ trace, this would be an enormous quantity of data. For example $W \approx 2^{12}$ is in the mid-range of column counts among various "table clusters" used in SP1. For a 32-bit field such as KoalaBear, this translates into a total of $4 \times W \times \mathcal{N}_{row} = 2^{36}$ bytes, or 64 GB. This is twice the global VRAM of an Nvidia 5090 GPU. Clearly, keeping around

the padding is not feasible. In order to mitigate these costs, the SP1 proof system uses the following techniques:

1. Instead of using a batch multilinear PCS directly, we use the jagged polynomial commitment scheme, which is a sparse PCS designed for exactly the kind of sparsity inherent in the table layout as described above. This enables the prover and verifier PCS costs to depend only on the size of the unpadded data (and notably, the verifier PCS costs can be made independent of W). For the internal dense PCS, we use batch BaseFold, which we view as a multilinear PCS for committing to a single multilinear in `log_total_unpadded_trace_area` variables. The evaluation proof for a point $\eta = \eta_n || \eta_k$ consists of

2. For the sumchecks, the contributions of the padding rows are always easy to compute and don't scale with the amount of padding. The prover makes frequent use of this fact.

Important Note: The Hypercube proof system verifier can be described as an arithmetic circuit. The circuit is determined by the total trace area of the unpadded data, and the tables included in the segment. Notably, the circuit is *independent* of the individual tables' unpadded heights. Hence, the total number of verifier circuits can be easily limited to the $\approx 2^{18}$ range, which makes it feasible to build them all instead of using a VM for recursion.

In the following subsections, we describe each section of the protocol in a bit more detail, with an eye towards making the soundness analysis in soundcalc more transparent. Whenever a variable appears in `typewriter font`, it refers to a parameter in the "sp1.toml" file in the soundcalc repository.

5.1 LogUp + GKR Details

TODO: Ask Min about the LogUp+GKR business.

5.2 Zerocheck Details

Recall that we have a trace of height $\mathcal{N}_{row}$ and width W on which we place `num_constraints` algebraic constraints, denoted $C_0, C_1, \ldots, C_{num_constraints}$. Each C_i is a polynomial of total degree at most 3 in W variables, though almost always it depends trivially on most of the variables. It is useful also to write $W = w_1 + \ldots + w_k$, where k is the number of tables in the cluster, and to group the constraints by table, so that $C_{i,j}$ refers to the j-th constraint on the i-th table. Let us write c_i for the number of constraints in the i-th table, and $p_{i,\ell}$ for the ℓ-th column of the i-th table. Finally, let us write h_i for the height of the i-th table (not including padding). The claim for our zerocheck is that, $\forall x \in \{0,1\}^{n_{row}}, \quad i \in [k], \quad j \in [c_i]$:

$$C_{i,j}(p_{i,1}[x], \ldots, p_{i,w_i}[x]) = \begin{cases} 0, & x < h_i, \\ C(0, \ldots, 0), & x \geq h_i \end{cases} \tag{16}$$

Note that the constraint evaluation on zero-padding may not result in 0, so at the moment our protocol does not match exactly with the usual zerocheck. However, note the following. If $\mathrm{geq}_{h_i}(x)$ is the function which, for $x \in \{0,1\}^{n_{row}}$, is zero if $x < h_i$ and 1 if $x \geq h_i$, then its multilinear extension $\widehat{\mathrm{geq}}_{h_i}(z)$ is relatively straightforward to evaluate at arbitrary $z \in \mathbb{F}^{n_{row}}$, using the following recursion relation. Write $z = (z_0, z_{rest}) \in \mathbb{F} \times \mathbb{F}^{n_{row}-1}$, and write $h_i = h_{i,0} * 2^{n_{row}-1} + h_{i,rest}$. Then, we have:

$$\widehat{\mathrm{geq}}_{h_i}(z) = z_0 \cdot (1 - h_{i,0}) + \widehat{\mathrm{eq}}(z_0, h_{i,0}) \cdot \widehat{\mathrm{geq}}_{h_{i,rest}}(z_{rest})$$

To prove this, first, we sample a random $\zeta \in \mathbb{F}^{n_{row}}$, $\lambda \in \mathbb{F}, \alpha \in \mathbb{F}$, and turn this collection of claims into the single sumcheck claim:

$$\sum_{i=0}^{k-1} \sum_{j=0}^{c_i-1} \lambda^i \alpha^j \widehat{\mathrm{geq}}_{h_i}(\zeta) = \sum_{x \in \{0,1\}^{n_{row}}} \widehat{\mathrm{eq}}_\zeta(x) \sum_{i=0}^{k-1} \lambda^i \sum_{j=0}^{c_i-1} \alpha^j C_{i,j}(p_{i,1}[x], \ldots, p_{i,w_i}[x]) \tag{17}$$